Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. For which step of generic sequential probabilistic inference must we know the measured system output z_k?

1 / 1 point

- ( ) 1a: State prediction time update.
- ( ) 1b: Error covariance time update.
- ( ) 1c: Predict system output.
- ( ) 2a: Estimator gain matrix.
- (•) 2b: State estimate measurement update.
- ( ) 2c: Error covariance measurement update.

Correct
Yes. We need to know z_k to perform the computations of this step.

2. Which steps of generic sequential probabilistic inference use the Kalman gain matrix L_k? (Select all that apply.)

1 / 1 point

- [ ] 1a: State prediction time update.
- [ ] 1b: Error covariance time update.
- [ ] 1c: Predict system output.
- [ ] 2a: Estimator gain matrix.
- [x] 2b: State estimate measurement update.

Correct
Yes. We need to know L_k to update the state prediction to make the state estimate.

- [x] 2c: Error covariance measurement update.

Correct
Yes. We need to know L_k to update the error covariance of the prediction to the error covariance of the estimate.

3. Which step of generic probabilistic inference requires the covariance of the measurement error $\Sigma_{\tilde{z},k}$?

1 / 1 point

- ( ) 1a: State prediction time update.
- ( ) 1b: Error covariance time update.
- ( ) 1c: Predict system output.
- (•) 2a: Estimator gain matrix.
- ( ) 2b: State estimate measurement update.
- ( ) 2c: Error covariance measurement update.

Correct
Yes. We need to know $\Sigma_{\tilde{z},k}$ to compute L_k.